JAMES MITCHELL

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PROFESSIONAL SUMMARY

I'm a curious Research Engineer with 2 years of hands-on experience in machine learning and data systems. Started my career building recommendation algorithms and have since moved into optimizing data pipelines at scale. I enjoy tackling performance problems and writing code that actually gets used by real teams. I'm based in Manchester but open to learning new approaches and technologies wherever they come from.

WORK EXPERIENCE

Research Engineer

DataFlow Systems

Manchester, United Kingdom

www.dataflowsystems.co

June 2023 - Present

Developed and optimized recommendation algorithms for e-commerce platform serving 2 million monthly active users

Reduced model inference latency by 35 percent through feature engineering and algorithm refinement

Conducted literature review on collaborative filtering techniques and implemented hybrid approach improving prediction accuracy by 12 percent

Collaborated with product team to establish A/B testing framework for validating model changes in production

Junior Machine Learning Engineer

Nexus Analytics Ltd

Manchester, United Kingdom

www.nexus-analytics.io

March 2022 - May 2023

Built data preprocessing pipelines in Python handling 500GB of daily transaction records using Apache Spark

Trained and evaluated multiple classification models for fraud detection achieving 94 percent precision on validation set

Documented model architecture and performance metrics for stakeholder presentations and internal knowledge base

Participated in code reviews and contributed to open source data validation library used internally

Data Science Intern

TechVision Research

Manchester, United Kingdom

www.techvision-research.net

September 2021 - February 2022

Created Python scripts to extract and clean data from various APIs and database sources for exploratory analysis

Assisted senior researcher in building time series forecasting models for demand prediction

Generated weekly reports with visualizations highlighting key trends and anomalies in datasets

Shadowed machine learning engineers during model deployment and learned containerization with Docker

EDUCATION

University of Manchester

MSc Computer Science with Data Science Specialization

Graduated: September 2021

Dissertation: Scalable Anomaly Detection in Time Series Data Using Autoencoders

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost

Data Engineering: Apache Spark, Pandas, NumPy, SQL

Cloud Platforms: AWS (EC2, S3, SageMaker), Google Cloud Platform

Tools and Frameworks: Git, Docker, Jupyter Notebook, Linux

Databases: PostgreSQL, MongoDB, Redis

GITHUB REPOSITORIES

recommendation-engine

https://github.com/jamesmitchell92/recommendation-engine

Python implementation of collaborative filtering algorithms with comparative performance analysis on MovieLens dataset

fraud-detection-pipeline

https://github.com/jamesmitchell92/fraud-detection-pipeline

End-to-end machine learning pipeline for transaction fraud classification with model serving API

time-series-anomaly-detection

https://github.com/jamesmitchell92/time-series-anomaly-detection

Autoencoder-based approach for detecting anomalies in multivariate time series data with visualization tools

data-validation-utils

https://github.com/jamesmitchell92/data-validation-utils

Lightweight Python library for schema validation and data quality checks in ETL workflows